Winston Liang

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Education

University of Toronto - Hong Kong Scholar, Dean's List Scholar — Sept 2024 – Expected 2028
Bachelor of Science, Specialist in Computer Science and Data Science (cGPA: 3.95/4.0) — *Toronto, CA*

Experience

Incoming AI Research Summer Intern — Starting May 2026
Multi-Agent Robotic Motion Laboratory @National University of Singapore — *Singapore*

Early Identification Program — Mar 2026 - Apr 2026
Selected as 1/163 students in North America @Citibank — *Remote, US*

Summer Intern — Jul 2025 – Aug 2025
Hong Kong Digital Policy Office — *Hong Kong SAR*

- Reduced vulnerability intelligence reporting time from 2+ hours to less than 1 minute (99.9% faster) by developing a Python scraping and alerting pipeline for public security advisories, enabling timely distribution of high-risk alerts to 70+ government bureaus and departments.
- Built a local model inference prototype using Ollama and Open WebUI, achieving ∼10 tokens/sec generation and 5 concurrent users on an Apple M3 Silicon, enabling staff to use web search feature with internal documents without sending sensitive information to external cloud services.
- Optimized on-premise model deployment for constrained hardware by evaluating quantization, model distillation, and retrieval pipelines, enabling efficient inference of 3B–32B parameter models for a 10+ user internal pilot system.
- Researched and tested approaches for improving reliability of model-generated outputs, including structured context retrieval and prompt guardrails, assessing their impact on reducing hallucinated responses and limiting sensitive information exposure in a prototype internal knowledge assistant.

Projects

SceneClarity - UTMIST ML Developer | *Pytorch, Tensorflow, Google Colab* — Sept 2025 – Present

- Developed a modular ML evaluation pipeline within the University of Toronto Machine Intelligence Student Team, achieved a reliability scoring system that assesses autonomous-vehicle perception robustness under adverse conditions.
- Applied transfer learning with ResNet-18 and ResNet-50 in PyTorch on the WeatherNet dataset (18,039 images), achieving 89.48% accuracy and 0.897 macro-F1 with ResNet-50.
- Collaborated with 8 people in an Agile workflow, managing tasks in Jira, contributing to pull-request reviews, design discussions, and weekly standups to maintain code quality and architectural consistency.

Scoliosis AI | *Next.js, PostgreSQL, AWS, Clerk, Prisma, NeonDB, PyTorch, YOLOv8* — Dec 2025 - Present

- Built an end-to-end full-stack AI web application for scoliosis detection from spinal X-ray images using YOLOv8 object detection model trained on ∼1,500 labelled medical images, achieving 96% precision in prediction.
- Deployed a containerized inference service by packaging FastAPI + YOLOv8 with Docker on AWS EC2, enabling authenticated API inference with ∼0.24 s end-to-end latency per X-ray image.
- Implemented a scalable image processing pipeline using AWS S3 presigned URLs for direct browser uploads and PostgreSQL via Prisma for metadata storage, enabling secure handling and retrieval of medical scan data.

UofT Course LLM | *Next.js, ChromaDB, SQLite, FastAPI* — Apr 2026

- Built a RAG academic advising system using OpenAI text-embedding-3-small for semantic embeddings, ChromaDB for vector similarity search, and GPT-4o-mini for grounded response generation over verified course and policy data.
- Engineered a constraint-satisfaction timetable generator using backtracking and conflict pruning, returning ranked conflict-free 3-5 course schedules in $\tilde{6}$-16 ms median latency while accounting for class preferences.

Blackjack-RL | *FastAPI, React, Vercel, OpenAI Gymnasium* — Apr 2026

- Engineered a full-stack application with 5 RL algorithms and featuring a real-time training dashboard with configurable hyperparameters, enabling users to play blackjack against RL models and sharpen their probability intuition
- Benchmarked policy convergence across 1M-episode training runs, demonstrating that learned policies outperform hand-coded basic strategy by 12% in house edge reduction within a constrained action space.

Scholarships / Honors

University of Toronto International Merit Award — awarded for academic merit.
Dato Tong, Luis Charitable Foundation Limited Scholarship — selected as 1 of the 4 Hong Kong Scholars.
The Regents Participation Awards — for making a significant contribution to university student life.

Skills

Languages: Python, Java, C, Bash, JavaScript, SQL
Tools: Git/GitHub, Jira, Linux, Docker, Google AppScript, Amazon Web Services, Tableau, Supabase, Prisma
Certificates: Machine Learning Specialization (DeepLearningAI), Azure AI Fundamentals (Microsoft), Introduction to Cybersecurity (Cisco), Introduction to Cloud Computing and Software Engineering (IBM)